

Strong and Weak Entities

What is a Strong Entity?

Definition

A Strong Entity is an entity that can be uniquely identified by its own attributes and can exist independently of other entities.

It has its own primary key.

It does not depend on another entity for identification.

A strong entity:

- Has its own Primary Key
- Exists independently
- Can be uniquely identified
- Does not require another entity to exist , StudentID

Even if there are no courses, departments, or enrollments, students can still exist.

Student is a Strong Entity.

What is a Weak Entity?

Definition

A Weak Entity is an entity that cannot be uniquely identified by its own attributes alone and whose existence depends on another (strong) entity.

It always depends on a strong entity.

Suppose we have a company.

Employees:

EMPLOYEEID	NAME
101	Aadya
102	Arpit

Each employee has family members.

Dependents

NAME	RELATION
Riya	Sister
Riya	Mother
Aman	Brother

Can you uniquely identify a dependent using only the dependent's name?

No.

Many employees may have a dependent named "Riya."

Therefore:

Dependent is a Weak Entity.

Sometimes, an object only makes sense in the context of another object. Without the parent entity, these objects lose their identity.

Suppose a hotel has:

Room Number: 101

Another hotel also has:

Room Number: 101

Can room number alone identify the room?

No.

Need: HotelID + RoomNumber

Room depends on Hotel.

Room is a Weak Entity.

Key Relationships & Identifiers

An **Identifying Relationship** is the relationship through which a weak entity is identified using the primary key of its owner (strong entity).

A **Partial Key (also called a Discriminator)** is an attribute of a weak entity that uniquely identifies it only within the context of its parent strong entity. eg room number is uniquely identifying a room within a hotel, without hotel number room number doesn't uniquely identify it.

Without the employee, the dependent no longer makes sense.

Weak entities have existence dependency on strong entities.

A weak entity depends on a strong entity for both identification and existence.

If the owner entity is deleted, the weak entity typically loses its meaning and is also removed (often via cascading delete).

| Why Can't Weak Entities Have Their Own Primary Key?

A weak entity does not have enough attributes to uniquely identify itself globally. Its attributes are only unique within the context of the owning entity.

Weak entities exist because some real-world objects only make sense in relation to another entity and cannot be uniquely identified on their own.